

Software Requirements Specification (SRS)

UOS ASSISTANT – Intelligent University Voice Assistant

Project Manager: Dr. Muhammad Ilyas

Internal Advisor: Dr. Muhammad Saad Razzaq

Project Team: Muhammad Asmaad Saeed, Laiba Siraj, Fatima Zahid,
Areeba Rehman

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Project Manager's Signature: _____

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Definition of Terms, Acronyms and Abbreviations

Term	Description
ASR	Automatic Speech Recognition
STT	Speech-to-Text
TTS	Text-to-Speech
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
CABE	Call Assistant Backend (Java Spring Boot service)
pgvector	PostgreSQL extension for vector similarity search
API	Application Programming Interface
REST	Representational State Transfer
SSH	Secure Shell - protocol for remote server access
FTP / SFTP	File Transfer Protocol / SSH File Transfer Protocol
OTP	One-Time Password
VoIP	Voice over Internet Protocol
PSTN	Public Switched Telephone Network
TLS	Transport Layer Security
SRTP	Secure Real-Time Transport Protocol
GDPR	General Data Protection Regulation
PDPL	Pakistan Personal Data Protection Law
HIPAA	Health Insurance Portability and Accountability Act

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the detailed requirements for the UOS ASSISTANT. It outlines the system's purpose, features, operating environment, and design constraints to guide its development, implementation, and evaluation. The intended audience for this document includes project advisors, developers, system architects, testers, evaluators, and university stakeholders, as well as the project review and evaluation committee, who will use it as a reference for design validation, performance assessment, and future enhancements

1.2 Project Overview

The UOS Assistant is an intelligent, Urdu-speaking virtual assistant designed to automate inbound customer support for the University of Sargodha (UOS). The system processes voice queries through a Retrieval-Augmented Generation (RAG) pipeline: Google Speech-to-Text (STT) transcribes the caller's voice, an Agent Controller orchestrates query embedding using OpenAI's text-embedding-3-small model, retrieves relevant context from a PostgreSQL vector database (pgvector), constructs a structured prompt combining the question and retrieved context, sends it to OpenAI GPT-4o Mini for response generation, and delivers the response back to the caller via Google Text-to-Speech (TTS). The system supports natural Urdu, English, and hybrid Urdu-English communication, removing language barriers and providing 24/7 availability for university callers.

1.3 Scope

The system's included functionalities will be:

- I. Automatically answers inbound calls and interacts with callers in real time.
- II. Uses Google Speech-to-Text (STT) for accurate transcription with Urdu dialect support, and OpenAI GPT-4o Mini for context-aware response generation.
- III. Supports bilingual communication in both English and Urdu, including hybrid Urdu-English code-switching.
- IV. Retrieves essential information from the PostgreSQL vector database (admission_document table), including admission schedules, fee structures, eligibility criteria, departmental contacts, office timings, and event details.
- V. Allows seamless call transfer to a human operator for complex or unresolved queries.

The system's excluded functionalities will be:

- I. Outbound or promotional calling.
- II. Video or chat-based communication.
- III. Integration with social media or non-telephony communication platforms.
- IV. External support for non-verbal communication modes (e.g., emails or text chat).

2. Overall System Description

2.1 User Characteristics

Users of the *UOS ASSISTANT* include a diverse range of individuals and groups. Primary users are students, faculty members, and administrative staff who contact the university for information regarding admissions, fee structures, schedules, departments, or general queries. Additionally, external callers such as parents, prospective students, and the general public may use the system to obtain information or make inquiries. Administrators and technical staff have elevated privileges to monitor performance, manage server configurations via SSH (PuTTY), inspect the PostgreSQL database using DBeaver, transfer data files using FileZilla, and ensure smooth operation. Stakeholders, including university management and project supervisors, oversee system performance, data insights, and alignment with institutional goals.

2.2 Operating Environment

The system operates on a Linux (Ubuntu) server hosted on Microsoft Azure cloud infrastructure. The PostgreSQL vector database (with pgvector extension) and all backend microservices run inside Docker containers for portability and isolation. The system is integrated with telephony services through VoIP/PSTN gateways. The development and operations team accesses the server remotely via SSH using PuTTY, transfers files using FileZilla FTP, and manages the database through DBeaver.

2.3 System Constraints

- It must comply with data protection regulations, including GDPR, PDPL, and HIPAA standards.
- The system is limited to inbound voice-based communication only and does not support outbound or promotional calling.
- It does not support video or chat-based communication interfaces.

- The system does not integrate with social media platforms or any non-telephony communication channels.
- It excludes external support for non-verbal communication modes, such as emails or text-based chat systems.
- The system requires a stable internet connection for real-time processing and API calls to OpenAI and Google Cloud services.
- The system must support Urdu-English code-switching and handle dialect variations commonly used in Pakistan.
- OpenAI API access (for embeddings and GPT-4o Mini) and Google Cloud API access (for STT and TTS) are required dependencies.

3. External Interface Requirements

3.1 Hardware Interfaces

- The Linux server (hosted on Microsoft Azure) must be capable of supporting simultaneous audio streaming, embedding generation, vector search queries, and real-time communication without performance degradation.
- The system is compatible with VoIP/PSTN telephony hardware, allowing inbound calls to be received through telephone lines, mobile phones, and softphone applications.
- It supports use with standard microphones, headsets, and desktop or laptop computers for voice input and output via the UOS Assistant Android application.
- No specialized or proprietary hardware components are required for system operation.
- For optimal performance, the deployment server should include multi-core processors, 8 GB or more RAM, sufficient storage for the PostgreSQL vector database, and reliable network connectivity to ensure smooth call handling and low-latency API communication.

3.2 Software Interfaces

- The system integrates with the Google Cloud Speech-to-Text (STT) API for converting spoken Urdu/English input into text with dialect support.
- It integrates with the Google Cloud Text-to-Speech (TTS) API for generating natural Urdu and English voice responses.
- It uses the OpenAI API for two purposes: text-embedding-3-small for generating 1536-dimensional vector embeddings of queries and documents, and GPT-4o Mini for context-aware response generation.

- The PostgreSQL database (with pgvector extension) serves as the vector store and is accessed by the Agent Controller for similarity search retrieval.
- The system communicates between modules using RESTful APIs (FastAPI-based Python backend and Java Spring Boot CABB).
- PuTTY is used as the SSH client for secure remote access to the Linux server.
- FileZilla is used as the FTP/SFTP client for uploading data files and downloading logs.
- DBeaver is used as the database client for inspecting and managing the PostgreSQL vector database.
- Docker is used for containerization of the PostgreSQL vector database and backend microservices.
- The system includes an urgency detection module to identify high-priority or time-sensitive queries and ensure appropriate routing or escalation.

3.3 Communication Interfaces

- All inter-service communication uses RESTful APIs over HTTPS for secure data exchange between the UI, Agent Controller, CABB, and external APIs.
- Server administration uses the SSH protocol (via PuTTY) over port 22 for secure remote access.
- File transfers between development machines and the server use SFTP (via FileZilla) over the SSH channel.
- Database connections to PostgreSQL are tunneled over SSH when accessed externally via DBeaver.
- All communication channels are secured using TLS (Transport Layer Security) and SRTP (Secure Real-Time Transport Protocol) to maintain data integrity, confidentiality, and protection against interception.
- Telephony communication uses VoIP/PSTN gateways managed by the Call Assistant Backend (CABB).

4. Functional Requirements

1. The system shall transcribe the caller's voice input into text using Google Speech-to-Text with Urdu dialect support.
2. The system shall route the transcribed query to the Agent Controller for processing.
3. The system shall generate a 1536-dimensional vector embedding of the user's query using OpenAI text-embedding-3-small.

4. The system shall perform vector similarity search on the PostgreSQL *admission_document* table to retrieve the top-K most relevant context chunks.
5. The system shall construct a structured LLM input by combining the system prompt, user question, and retrieved context.
6. The system shall send the constructed input to OpenAI GPT-4o Mini and receive a structured JSON response containing the *response_text*.
7. The system shall synthesize the *response_text* into natural voice output using Google TTS in the caller's language.
8. The system shall support Urdu, English, and hybrid Urdu-English (code-switching) communication.
9. The system shall retrieve information from the university database, including admissions, fee structures, eligibility criteria, courses, and departmental contacts.
10. The system shall transfer complex or unresolved queries to a human operator via the CABA (Call Assistant Backend).
11. The system shall log all call sessions, transcriptions, retrieval results, and LLM responses for monitoring and continuous improvement.
12. The system shall support an offline RAG ingestion pipeline that processes CSV/PDF data files, generates embeddings, and stores them in the PostgreSQL vector store.

5. Non-Functional Requirements

5.1 Performance Requirements

- The system must respond to user voice input within 2 seconds end-to-end (from speech input to voice response) to ensure real-time interaction.
- The vector similarity search on PostgreSQL pgvector should return top-K results within 200 milliseconds for typical query loads.
- OpenAI API calls (embedding and GPT-4o Mini) should complete within 1 second under normal network conditions.
- The system should maintain an uptime of at least 99% for consistent service availability.
- The system should support concurrent handling of multiple inbound calls without performance degradation.

5.2 Safety Requirements

- The system must ensure no data loss during call handling, even in the event of a network interruption or system failure.
- All critical operations should include data backup and recovery mechanisms to prevent loss of information.
- The PostgreSQL vector database must be backed up regularly, with automated snapshots stored on Azure cloud storage.
- Call session logs and response history must be persisted to disk for audit and recovery purposes.

5.3 Security Requirements

- The system must implement end-to-end encryption (TLS) for all API communications and strict access control to protect sensitive information.
- Server access must use SSH key-based authentication via PuTTY - password-based SSH login should be disabled.
- All database credentials, API keys (OpenAI, Google Cloud), and secrets must be stored in environment variables, never hardcoded.
- Access to DBEaver and FileZilla connections must require strong authentication and be limited to authorized team members only.
- The system must comply with GDPR, PDPL, and HIPAA standards to ensure user data privacy and legal compliance.
- The system will not use biometric or OTP-based authentication methods to maintain simplicity and avoid storing sensitive identifiers.
- Voice recordings and transcriptions must be encrypted at rest and only retained for the duration necessary for analytics and system improvement.

5.4 User Documentation

- A comprehensive User Manual, Administrator Guide, and Troubleshooting Documentation will be provided with the system deployment.
- These documents will assist end-users and technical staff in installation, operation, maintenance, and issue resolution.

6. Assumptions and Dependencies

- The system assumes continuous and stable internet connectivity for seamless operation of API calls to OpenAI and Google Cloud.
- It assumes accurate transcription performance from Google STT for reliable Urdu and English speech processing.

- The system depends on the reliability and availability of external cloud APIs: OpenAI (embeddings + GPT-4o Mini) and Google Cloud (STT + TTS).
- The system depends on the PostgreSQL database with pgvector extension being correctly installed, configured, and populated with university data via the RAG ingestion pipeline.
- It depends on the Microsoft Azure cloud platform for hosting the Linux server, with Docker available for containerization.
- It depends on the team having proper SSH credentials for PuTTY access, FTP credentials for FileZilla, and database credentials for DBeaver.
- The system depends on regular maintenance and updates of all software components, Docker images, and external integrations to ensure consistent performance.

7. References

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